

Seller Churn: Findings & Intervention Plan

Marketplace Seller Health Program · Analysis of 6,000 simulated sellers

41.1%

Sellers churn within 12 months

27.2%

of sellers flagged High Risk today

0.80 AUC

early-warning model accuracy

~390%

estimated ROI on targeted intervention

What's Driving Churn

Churn is not random — it's concentrated in sellers with declining order momentum, high return rates, and repeated fulfillment (SLA) failures. The strongest signals, ranked by predictive strength in the model:

- **Order velocity decline** — a drop in 30-day order volume is the clearest early warning sign; sellers show slowing activity well before they formally stop selling.
- **Return rate** — sellers above the 75th percentile return rate churn at roughly double the base rate, concentrated in Fashion and Mobiles & Accessories.
- **SLA / fulfillment breaches** — logistics failures compound with returns, and are most severe for Tier-3 town sellers (49.4% churn vs. 36.7% for Tier-1 metro sellers).
- **Rating decline & support ticket volume** — a downward 90-day rating trend and rising support load add incremental signal on top of the operational metrics above.
- **Category effect** — Fashion has the highest churn rate (51.8%), Grocery the lowest (34.0%), reflecting structural differences in return burden and repeat-purchase behavior.

Seller Archetypes

Behavioral clustering (KMeans on operational metrics, independent of the churn label) surfaces four recurring seller archetypes:

Archetype	Sellers	Churn Rate	Avg. Mo. Revenue	Read
Struggling / High-Return Risk	1,335	77.8%	Rs.33,316	Urgent — highest priority for intervention
Declining Momentum	1,397	44.3%	Rs.34,295	Early warning stage — velocity/rating slipping
Steady Mid-Tier	1,372	38.0%	Rs.36,010	Stable but not growing — coaching upside
Star Performers	1,896	15.0%	Rs.57,705	Healthy — retain via recognition, not coaching

Recommended Interventions

Driver Signal	Intervention	Owner
High return rate	Inventory/QC coaching: listing accuracy, packaging audit, sizing guides (Fashion)	Seller Success
SLA breaches	Logistics support: fulfillment partner enrollment, dispatch SLA training	Logistics Ops
Weak pricing competitiveness	Automated pricing coaching / repricing tool recommendation	Growth/Pricing
Negative order velocity	Demand boost: ad credits, category placement, promo eligibility	Marketing
Declining ratings / rising tickets	Proactive CX outreach + root-cause review call	Support

Prioritization & Expected Impact

1,631 sellers (27.2% of the base) are flagged **High Risk** by the churn model today. Targeting this group with the interventions above, assuming a 25% save rate, a 4-month expected lifetime without intervention vs. 14 months if retained, and Rs.2,000 average coaching cost per seller:

Sellers targeted	1,631
Expected sellers saved (@25% success)	~407
Avg. incremental LTV per saved seller	Rs.39,313
Total incremental LTV	Rs.1.60 crore (Rs.16.0M)
Total program cost	Rs.32.6 lakh (Rs.3.26M)
Estimated ROI	~390%

Next steps: (1) Pilot interventions on the top 200 highest-risk sellers for one quarter to validate the assumed 25% save rate before scaling. (2) Route the weekly at-risk feed from the model directly to Seller Success/Logistics/Pricing owners per the table above. (3) Re-run the model monthly and track precision/recall against actual outcomes to recalibrate thresholds.

Source: analysis.ipynb (EDA, driver analysis, KMeans segmentation, Logistic Regression / Random Forest / Gradient Boosting churn models). Data is simulated for portfolio purposes; recalibrate commission rate, seller lifetime, and coaching cost assumptions against real platform economics before use.